4.1 Answer each of the following:

a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2.

b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.

c. Suppose that a simple regression has quantities $\sum(y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2.

$\Sigma y_i = \bar{y}N = 19.21 \times 20$.

(a) $R^2 = \frac{SSR}{SST}$.

$SST = \Sigma(y_i - \bar{y})^2 = \Sigma(y_i^2 - 2y_i\bar{y} + \bar{y}^2) = \Sigma y_i^2 - 2\bar{y}\Sigma y_i + \Sigma\bar{y}^2 = \Sigma y_i^2 - 2\bar{y}\cdot N\cdot\bar{y} + \Sigma\bar{y}^2 + N\bar{y}^2 = \Sigma y_i^2 - N\bar{y}^2$

$= 7825.94 - 20 \times (19.21)^2 = 445.458 \Rightarrow R^2 = \frac{375.47}{445.458} = 0.8429$ #

(b) $R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{SSE}{725.94} = 0.7911 \Rightarrow SSE = 151.6489$, $\hat{\sigma}^2 = \frac{\Sigma\hat{e}_i^2}{N-2} = \frac{SSE}{N-2} = \frac{151.6489}{18} = 8.4249$ #

(c) $\Sigma(y_i - \bar{y})^2 = 631.63 = SST$, $\Sigma\hat{e}_i^2 = 182.85 = SSE \Rightarrow R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.7105$ #

4.3 We have five observations on x and y. They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^{5}\hat{e}_i^2 = 3.6$, $\sum_{i=1}^{5}(x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^{5}(y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

a. the predicted value of y for $x_0 = 4$.

b. the $\text{se}(f)$ corresponding to part (a).

c. a 95% prediction interval for y given $x_0 = 4$.

d. a 99% prediction interval for y given $x_0 = 4$.

e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

$SSE = 3.6$

$SST = 10$

$SSR = 10 - 3.6 = 6.4$

(a) $\hat{y}_i = 1.2 + 0.8 \times 4 = 1.2 + 3.2 = 4.4$ #

(b) $\hat{\sigma}^2 = \frac{\Sigma\hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = 1.2$, $\bar{x} = \frac{3+2+1-1+0}{5} = 1$

$se(f) = \sqrt{\hat{\sigma}^2\left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\Sigma(x_i - \bar{x})^2}\right]} = \sqrt{1.2\left(1 + \frac{1}{5} + \frac{(4-1)^2}{10}\right)} = 1.5875$ #

(c) $t_{3;0.025} = 3.182$, 95% P.I. $= \hat{y}_i \pm t_{3;0.025}\, se(f) = 4.4 \pm 3.182 \times 1.5875 = (-0.6514, 9.4514)$ #

(d) $t_{3,0.005} = 5.841$, 99% P.I. $= 4.4 \pm 5.841 \times 1.5875 = (-4.8726, 13.6726)$ #

(e) $x = \bar{x} = 1$. $\hat{y}_i = 1.2 + 0.8 = 2$. $se(f) = \sqrt{1.2\left(1 + \frac{1}{5} + 0\right)} = 1.2$

$t_{3,0.025} = 3.182$, 95% P.I. $= 2 \pm 3.182 \times 1.2 = (-1.8184, 5.8184)$.

the width of (e) $= 5.8184 - (-1.8184) = 7.6368$

the width of (c) $= 9.4514 - (-0.6514) = 10.1028 \Rightarrow$ the width of (c) larger than (e) #

When the x_0 closed to $\bar{x}$ → $se(f)$ smaller → the uncertainty of prediction will decline. #

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.

```
> summary(model1)

Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.32528 -0.19199  0.00122  0.21678  0.86609 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,	Adjusted R-squared:  0.5408 
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16

> slope_a
[1] 9.019661
> elas_a
[1] 0.9833701
```

(a) A one-unit increase in square footage is associated with an approximate 3.60% increase in house price. At the sample means, the marginal effect (slope) is 9.0197, and the price elasticity with respect to square footage is 0.9834.

b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.

```
> summary(model2)

Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36864 -0.20849  0.00487  0.23204  1.21099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.04965    0.15797   12.97   <2e-16 ***
log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,	Adjusted R-squared:  0.4728 
F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16

> slope_b
[1] 9.399923
> elas_b
[1] 1.024828
```

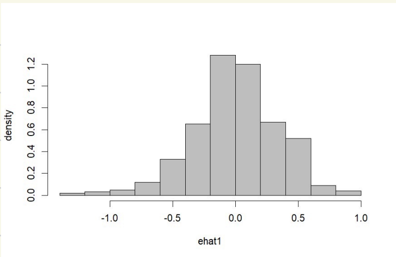
We estimate that the elasticity is 1.025: a 1% increase in square footage is estimated to increase the house price by 1.025%. Furthermore, at the sample means, each additional square foot is estimated to increase the house price by approximately 9.40.

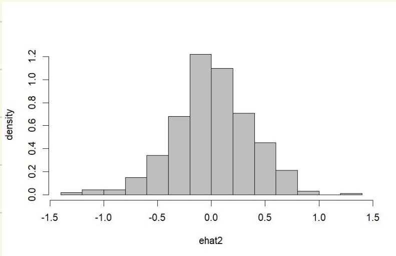
c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the "generalized" R^2 measure for the models in (b) and (c).

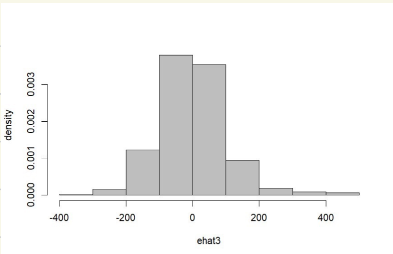
```
> rsq_3
[1] 0.6413167
> rg_1
[1] 0.6621612
> rg_2
[1] 0.6445084
```

(c) Based on the results, the generalized R^2 values are ranked as follows: Log-linear model (0.662) > Log-log model (0.644) > Linear model (0.641). Since the log-linear model yields the highest generalized R^2, we conclude that it provides the best fit to the data among the three models.

d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?



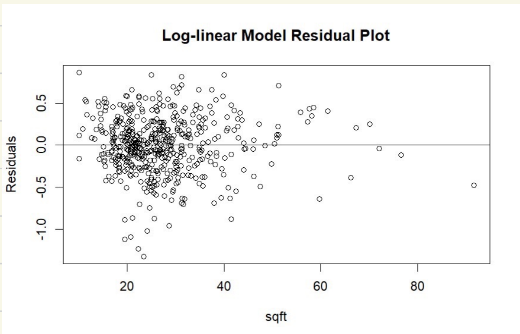


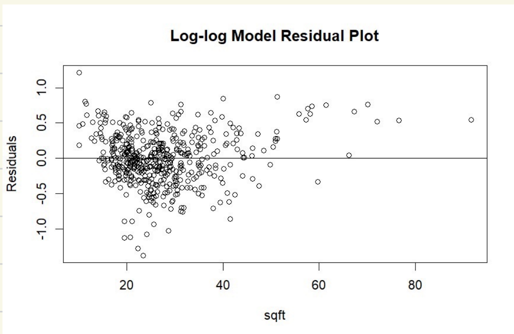


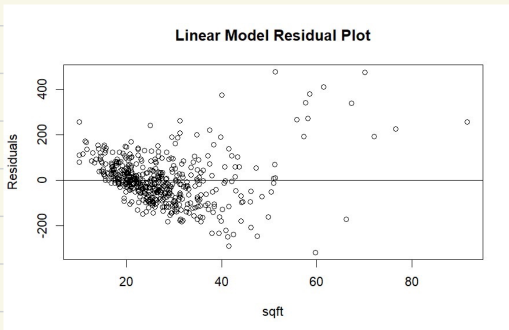

```
> jarque.bera.test(ehat1)

        Jarque Bera Test

data:  ehat1
X-squared = 26.679, df = 2, p-value = 1.61e-06

> jarque.bera.test(ehat2)

        Jarque Bera Test

data:  ehat2
X-squared = 14.279, df = 2, p-value = 0.0007933

> jarque.bera.test(ehat3)

        Jarque Bera Test

data:  ehat3
X-squared = 221.11, df = 2, p-value < 2.2e-16
```

(d) The Jarque–Bera tests for all three models yield p-values far below 0.05, leading to the rejection of the null hypothesis of normality for the residuals. While all models fail the test, the log-log model's distribution (ehat2) appears relatively more symmetric and has the smallest X-squared, suggesting it is the closest to normality among the three.

e. For each of the models in (a)–(c), plot the least squares residuals against *SQFT*. Do you observe any patterns?







(e) The linear model exhibits a clear funnel-shaped pattern, indicating that the error variance increases as SQFT increases (a violation of the homoskedasticity assumption). In contrast, the residuals for both the log-linear and log-log models appear much more randomly dispersed around the zero line. This demonstrates that applying logarithmic transformations effectively mitigates the heteroskedasticity issue and helps stabilize the variance of the residuals.

f. For each model in (a)–(c), predict the value of a house with 2700 square feet.

```
> pred_1
       1
226.9817
> pred_2
       1
243.1514
> pred_3
       1
246.4557
```

(f) the linear model predicts a price of 226.98 (in thousands of dollars), the log-linear model predicts 243.15, and the log-log model yields 246.46. These results demonstrate that all three models provide reasonable and relatively close estimates, ranging from approximately \$226,980 to \$246,460 for a 2,700-square-foot house.

g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.

```
> PI_1
       fit      lwr      upr
1 214.2336 109.7685 418.1167
> PI_2
       fit      lwr      upr
1 227.5386 111.1406 465.8406
> PI_3
       fit      lwr      upr
1 246.4557 44.27727 448.6341
```

(g) the 95% prediction intervals for the house price are [109.77, 418.12] for the log–linear model, [111.14, 465.84] for the log–log model, and [44.28, 448.63] for the linear model. While all three models provide wide prediction intervals due to the large variance in house prices, the linear model yields a much lower and less realistic lower bound ($44.28) compared to the two logarithmic models. This suggests that the log transformations help prevent extreme or nonsensical predictions at the lower end.

h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

I choose the log–linear model over the linear model for four main reasons:

1. Best Fit: It has the highest generalized R^2(0.662).
2. Improved Normality: The log transformation significantly reduces the Jarque–Bera statistic, bringing residuals closer to normal.
3. Constant Variance: It successfully eliminates the funnel–shaped heteroskedasticity seen in the linear model.
4. Realistic Forecasts: It provides a much more reasonable lower bound for the 95% prediction interval.

In short, the log–linear model perfectly resolves the severe violations of assumptions in the linear model and gives the most reliable estimates.